

ELECTRICAL RESEARCH AND DEVELOPMENT ASSOCIATION

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TEST REPORT		SHEET 1 of 4	
NAME & ADDRESS OF CUSTOMER Savita Polymers Ltd., 66/67, Nariman Bhavan, 6 th Floor, Nariman Point, Mumbai.	REPORT No.	: RP-1617-028800	
	DATE	: 21-09-16	
	CUSTOMER REF. No.	: BTHF/Erda/1	
	DATED	: 12-09-16	
	DATE OF SAMPLE RECEIPT	DATE OF TESTING	
	16-09-16	16-09-16 to 21-09-16	
SAMPLE DESCRIPTION Unused Natural Ester Fluid, BIOTRANSOL HF.	SAMPLE IDENTIFICATION ERDA SAMPLE CODE : ERDA-00160420		
TEST DETAILS Test details on sheet 2 of 4	TEST SPECIFICATION: IEC:62770-2013		
Remarks: The sample conforms to the requirements of all the above mentioned tests.			
 PREPARED BY		 CHECKED BY	
		 (Dr. M.J. Patel) APPROVED BY	
Note: 1. This report relates only to the particular sample received for testing in good condition at ERDA. 2. This report cannot be reproduced in part under any circumstances. 3. Publication of this report requires prior permission in writing from Director, ERDA. 4. Only the tests asked for by the customer have been carried out. 5. In case of dispute Vadodara will be the exclusive jurisdiction & shall be construed as Where the cause has arised. caution ERDA is not responsible for the authenticity of photocopied or reproduced test reports. ERDA provides support to customers for verification of the authenticity of test reports issued by ERDA. 			

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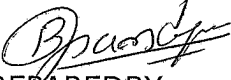
DATE: 21-09-16

SHEET 2 of 4

TEST DETAILS:

TEST SPECIFICATION: IEC 62770-2013 &

Sr. No	Test particulars	Clause Nos. of IEC 62770-2013
Physical		
1.	Appearance	4.2.1
2.	Kinematic Viscosity	4.2.2
3.	Pour Point	4.2.3
4.	Water Content	4.2.4
5.	Density	4.2.5
Electrical		
6.	Breakdown Voltage	4.3.1
7.	Dielectric Dissipation Factor	4.3.2
Chemical		
8.	Corrosive Sulphur	4.4.2
Performance		
9.	Oxidation Stability	4.5.1
Health, Safety and Environment (HSE)		
10.	Fire Point	4.6.1
11.	Flash Point	4.6.1
12.	Polychlorinated Biphenyls (PCBs)	4.6.2


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SHEET 3 of 4

Sr. No.	Cl. No.	Test particulars	Requirement as per specification	Obtained value	Remarks
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RESULTS :

Physical

1.	4.2.1	Appearance	Clear, free from sediment and suspended matter	Clear, free from sediment and suspended matter	Conforms
2.	4.2.2	Kinematic Viscosity, mm ² /s			
		- at 100°C	Max. 15	8.3136	Conforms
		- at 40°C	Max. 50	36.1420	Conforms
3.	4.2.3	Pour Point, °C	Max. -10	-18	Conforms
4.	4.2.4	Water Content, mg/kg	Max. 200	68	Conforms
5.	4.2.5	Density at 20°C, kg/m ³	Max. 1000	922	Conforms

Electrical

6.	4.3.1	Breakdown voltage, kV	Min. 35	89	Conforms
		a) Individual breakdown Voltages, kV: 90.3,87.4,87.0,89.9,85.8,91.4			
		b) Type of electrodes: Spherical			
		c) Frequency of the test voltage: 50 Hz			
		d) Oil temperature: 26.5°C			
7.	4.3.2	Dielectric Dissipation factor at 90°C	Max. 0.05	0.019	Conforms

Chemical

8.	4.4.2	Corrosive Sulphur at 150°C (for 48hrs)	Non Corrosive	Non Corrosive	Conforms
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Sr. No.	Cl. No.	Test particulars	Requirement as per specification	Obtained value	Remarks
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RESULTS :

Performance

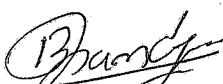
9.	4.5.1	Oxidation Stability at 120°C (48 Hrs.) Annex-A, After Oxidation			
	4.5.2	Total Acidity, mg KOH/gm oil	Max. 0.06	0.05	Conforms
	4.5.3	Kinematic Viscosity at 40°C, %	Max. 30% Increase Over Initial Value	9	Conforms
	4.5.4	Dielectric Dissipation factor at 90°C	Max. 0.5	0.083	Conforms

Health, Safety and Environment (HSE)

10.	4.6.1	Fire Point, °C	Min. 300	328	Conforms
11.	4.6.1	Flash Point, °C	Min. 250	258	Conforms
12.	4.6.2	Polychlorinated Biphenyls (PCBs), mg/kg	Max. 2	Not detected	Conforms

Test Conditions for Dielectric Dissipation factor:

- Type of Cell used: IEC : 60247 - 2004 (Fig. 1)
- Average Voltage gradient in the sample while under test: 250 V/mm
- Frequency of applied voltage: 50 Hz
- Temperature of the room during test: 26.4°C
- Humidity of the room during test: 38%


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